Response to the Reviewers

Title: Synthetic Data Generation for Joint Electric Vehicle Driving and Charging Events via Deep Generative Networks

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Overall Comments

Dear Zhibin,

Thank you for submitting your manuscript to Transportation Research Part C. I have completed my evaluation of your manuscript. The reviewers recommend reconsideration of your manuscript following major revision. I invite you to resubmit your manuscript after addressing the comments below. Please resubmit your revised manuscript by Oct 08, 2025.

When revising your manuscript, please consider all issues mentioned in the reviewers' comments carefully: please outline every change made in response to their comments and provide suitable rebuttals for any comments not addressed. Please note that your revised submission may need to be re-reviewed.

Transportation Research Part C values your contribution and I look forward to receiving your revised manuscript.

Kind regards,
Hai Yang
Editor
Transportation Research Part C

Messages From Authors

Dear Editor and Reviewers,

We sincerely thank you for your constructive feedback, which has significantly contributed to the enhancement of our manuscript. We have carefully addressed your comments and integrated your insightful suggestions into the revised manuscript. The key modifications made are outlined as follows:

- We introduced four additional evaluation metrics to more comprehensively assess model performance. Accordingly, the entire discussion of evaluation metrics has been revised in Section 3 of the revised manuscript. We have also updated the comparative analysis of different models, with corresponding results now presented in Tables 6, 7, 9, and 11, together with the revised discussions.
- In response to the reviewer's request for robustness analyses, we added three new subsections (Sections 4.5, 4.6, and 4.8). These sections evaluate: (i) the convergence efficiency of Gibbs sampling, (ii) the necessity of modeling temporal dependencies, and (iii) the robustness of the framework under different training set sizes.

We respond to the comments in an itemized manner. The statement for each comment is boxed, followed by our written responses. All the page and reference numbers mentioned in the reviewers' comments are kept intact based on the original manuscript, while those in our responses are

adapted as per the revised manuscript unless otherwise stated. For your convenience, the revised sections in the manuscript are highlighted in blue, and quotations from the manuscript are marked in red throughout this letter.

We would like to thank you for collecting and relaying the review comments to us, and hope that you find the revised manuscript satisfactory.

Sincerely,
Zhibin Chen

Response To Reviewer #1

Overall Comments

This paper proposes a deep generative neural network to synthesize joint electric vehicle (EV) driving and charging event data. The topic is timely given the growing adoption of EVs. However, this reviewer has the following concerns.

Response

We thank the reviewer for the positive assessment of the timeliness of our work. We appreciate the concerns raised and provide detailed point-by-point responses below.

Reviewer Comment 1.1

In Section 1, several existing methods for generating synthetic EV data are introduced. It is suggested to include a comparative table summarizing these works across various aspects, such as dataset features, timeframes, data generation methods, and their respective advantages and disadvantages compared to the proposed approach.

Response

We sincerely thank the reviewer for this insightful suggestion. In the revised manuscript, we have added a new comparative table (Table 1) on Page 3, which systematically summarizes representative studies on synthetic EV data generation. The table compares existing works across dataset

Table A1: Summary of studies on EV data synthesis

Literature	Synthetic Data	Method	Strength/Limitation
Wang et al. (2017)	EV travel chains	Naive Bayes method	Generates flexible travel chains but lacks detailed charging behavior.
Schäuble et al. (2017)	Aggregate charging load	Simulation-based approach	Scales to system-level planning but ignores individual-level behaviors.
Lahariya et al. (2020)	Charging demand	Gaussian mixture model	Captures station demand but overlooks individual charging patterns.
Einolander and Lahdelma (2022)	Charging events	Multivariate copula	Models joint event variables but not full driving–charging sequences.
Liu et al. (2022)	Driving and charging events	Activity-based simulation	Links activity patterns with charging but requires rich data and omits realistic event chains.
Razghandi et al. (2023)	Aggregate charging load	VAE-GAN hybrid	Produces realistic load profiles but does not capture driving–charging coupling.
Ge et al. (2024)	Charging station load	SeqGAN–LSTM	Learns spiky station load patterns but lacks individual EV representation.
Li et al. (2024a)	Charging events	Gibbs sampling + Deep neural networks	Generates multi-variable charging events but not integrated driving–charging chains.

characteristics, data types, methodological approaches, and highlights their strengths and limitations relative to our framework. For the reviewer's convenience, the table is also reproduced above (Table A1) in this response letter, in addition to its incorporation in the updated manuscript.

In addition, we have revised the related paragraph on Page 3 in the updated manuscript to explicitly introduce this table. The updated text is highlighted in red in the manuscript and reads as follows:

"For EV data synthesis, we provide a summary in Table 1, which categorizes existing studies by their methodological approaches, the types of EV data synthesized, and their respective advantages and limitations."

Reviewer Comment 1.2

> A key contribution of the proposed model is the integrated generation of both driving and charging events. However, the case study lacks comparisons with existing models that generate these separately. Including such comparative evaluations would strengthen the validation of this contribution.

Response

We thank the reviewer for raising this point. We would like to emphasize that the central contribution of our framework is precisely the *integrated generation of full individual driving–charging sequences*, rather than producing driving and charging events in isolation. In real-world mobility, driving and charging are inherently interdependent: whether the next event is a trip or a charging session will be affected by the preceding state of charge (SoC) and spatial location. This sequential dependency structure is crucial for faithfully capturing user behavior and cannot be represented by models that treat driving and charging as independent processes. Therefore, a direct comparison with such separate-generation approaches would not be meaningful for our task, as they fundamentally lack the ability to model sequence-level coherence that our framework is designed to address.

In the following, we further examined the necessity of explicitly modeling temporal dependencies through an ablation study, in which models were trained with different historical horizons, i.e., conditioning only on the most recent N events ($N \in \{1, 5, 10, 20, 30\}$). The resulting performance changes are summarized in Table A2. Please note that this table is evaluated using the revised set of metrics; we kindly refer the reviewer to Comment 1.6 for a detailed explanation of the updated evaluation criteria before interpreting these results. Table A2 shows that incorporating longer event histories (5–20 events) improves fidelity, diversity, and memorization compared with the 1-event Markov baseline. However, gains plateau beyond 20 events, suggesting that most useful dependencies are already captured within a moderate horizon. This aligns with real-world EV usage, where daily and weekly routines (e.g., commuting or evening charging) dominate, making medium-range dependencies both sufficient and effective.

In addition, we have added a sensitivity analysis on temporal dependency length in the revised manuscript. This new subsection is included on Page 33 for clarity. For the reviewer's convenience, we reproduce the newly added paragraph below. The corresponding results are reported in Ta-

Table A2: Comparison of results under different temporal dependency horizons

Horizon	Fidelity				Diversity		Memorization
	ρ_1	ρ_2	Joint-KL	Precision	Recall	Coverage	ρ_3
1 events	0.29	0.58	0.99	0.69	0.88	0.79	0.78
5 events	0.23	0.53	0.97	0.71	0.94	0.80	0.81
10 events	0.20	0.51	0.94	0.76	0.93	0.82	0.84
20 events	0.20	0.52	0.95	0.75	0.94	0.82	0.83
30 events	0.19	0.50	0.93	0.77	0.95	0.83	0.86

ble A2 (i.e., Table 8 in the revised manuscript), which remains the same as presented above.

"We further conducted a sensitivity analysis on the temporal dependency length to examine how much historical context is required for effective generation. Specifically, models were trained with horizons of 1, 5, 10, 20, and 30 preceding events (Table 8). The results show that extending the horizon length consistently improves fidelity and diversity: moving from a first-order Markov model (1 event) to a 10-event dependency significantly reduces joint errors and improves recall and coverage. However, beyond 20 events, the gains become marginal, and performance metrics approach a plateau.

This pattern can be explained by the inherent regularity in individual travel and charging behaviors. Most users follow relatively stable daily routines (e.g., commuting and overnight charging), so the critical dependencies can already be captured within a moderate horizon. Longer temporal windows add limited information, as distant events often provide redundant or weak signals. Overall, the sensitivity analysis confirms that while temporal dependency is critical for generating coherent charging–driving sequences, the model does not require excessively long histories, which supports tractability and scalability."

Reviewer Comment 1.3

> Three indicators are selected to assess the model's generative capabilities and privacy protection performance. Please clarify the rationale behind choosing these specific indicators and discuss whether other relevant metrics were considered.

Response

Thank you very much for the helpful suggestion. We first clarify the rationale and our metric choices as follows.

- $\boldsymbol{\rho_1}$ compares **univariate** marginals; uses Jensen–Shannon divergence for discrete variables and 1-Wasserstein distance for continuous variables—both are widely used for evaluating synthetic tabular data (e.g., Hudovernik et al., 2024; Shi et al., 2025; Stenger et al., 2024). These choices are symmetric, interpretable, robust to sparse categories.

- $\boldsymbol{\rho_2}$ evaluates **derived** and/or **conditioned** variables that capture joint relationships between multiple features. For instance, speed is derived as distance divided by driving duration, and charging power as the change in SoC over charging duration. We further stratify these derived variables by discrete attributes such as battery capacity, user label, and spatial index, thereby enabling the comparison of conditional distributions across meaningful subgroups. For each subgroup, we compute distributional distances using Jensen–Shannon divergence for discrete outcomes and 1-Wasserstein distance for continuous outcomes. In this way, ρ_2 transforms high-dimensional joint dependencies into physically interpretable variables that reflect both behavioral realism and subgroup consistency.

- $\boldsymbol{\rho_3}$ evaluates whether the generative model **memorizes** training data rather than learning generalizable patterns. The key idea is to use the **closest distance ratio** between training, validation, and synthetic samples as an indicator of potential leakage, a criterion that has become common in evaluating generative models (van den Burg and Williams, 2021; Fang et al., 2024; Shi et al., 2025). For each training instance, we compute its nearest-neighbor distance to both the validation set and the synthetic set. If the distance to synthetic samples is systematically smaller than that to validation samples, i.e., $\rho_3 > 1$, it suggests that the model may have overfit or memorized the training data. Conversely, $\rho_3 \approx 1$ indicates good generalization, while values below 1 may imply underfitting.

In addition, after carefully reviewing relevant literature and considering the feedback from Reviewer 2, we expand our evaluation framework by including several additional metrics, organized along three key aspects: fidelity, diversity, and memorization.

- **Fidelity:** measures how closely the generated data resemble the real data distribution. High fidelity means that synthetic samples are realistic and statistically consistent with the real dataset. In addition to our original metrics, i.e., ρ_1 and ρ_2, we incorporate the kNN–KL divergence estimator for mixed-type distributions (Ross, 2014; Gao et al., 2017; Wang et al., 2009), which captures conditional dependencies beyond marginal or derived variables. We did not adopt kernel density estimation (KDE)–based approaches (Moon et al., 2017), since they

typically suffer from the curse of dimensionality, and become computationally expensive for large-scale tabular datasets. We also adopt the kNN-based **precision** metric (Kynkäänniemi et al., 2019; Naeem et al., 2020; Cheema & Urner, 2023), which quantifies the proportion of generated samples lying within the support of the real data distribution. Compared with earlier approaches that defined precision using a fixed-radius ϵ ball (Sajjadi et al., 2018), the kNN formulation is both more robust to scaling, adaptive to heterogeneous data density, and computationally more scalable for large mixed-type tabular datasets. Together, these measures strengthen the fidelity assessment by combining global distributional similarity and local neighborhood realism.

- **Diversity:** evaluates how well the generated data capture the full variability of the real distribution, including rare modes and less frequent patterns. High diversity indicates that the generator avoids mode collapse and reproduces the spread of real-world behaviors. To this end, we include the kNN-based **recall** and **coverage** metrics (Naeem et al., 2020; Cheema & Urner, 2023). Recall quantifies the proportion of real samples supported by generated samples, while coverage emphasizes whether generated samples span distinct modes of the real distribution. These measures provide a richer characterization of diversity than ρ_1 or ρ_2 alone. We did not adopt KDE–based diversity measures (Moon et al., 2017) or vision-oriented scores such as Fréchet Inception Distance (Heusel et al., 2017), since they either suffer from scalability issues in high-dimensional tabular data or depend on pretrained feature extractors that are not well defined outside the image domain. In contrast, kNN-based recall and coverage are parameter-light, density-adaptive, and directly applicable to mixed-type datasets.

- **Memorization:** assesses whether the generative model has memorized training samples rather than learning generalizable patterns. A robust model should balance fidelity and diversity without leaking training data. In addition to our ρ_3 metric based on the closest distance ratio (van den Burg & Williams, 2021; Fang et al., 2024; Shi et al., 2025), we also reviewed alternative approaches for memorization assessment proposed in the literature, such as privacy risk audits and adversarial evaluation protocols (Stadler et al., 2022; Shokri et al., 2017; Hilprecht et al., 2019). These methods are primarily effective in domains such as image data synthesis and medical record sharing, where adversarial risk auditing and membership inference attacks have been extensively studied. In our case, we focus on tabular mixed-type EV data, where such setups are less practical and interpretable. We therefore retain ρ_3 as a practical and interpretable proxy for memorization risk.

We have updated our results and manuscript according to the new evaluation standards. Specifically, Table 6 (Page 25, updated), Table 7 (Page 26, updated), Table 9 (Page 34, updated), and Table 11 (Page 36, updated) have been revised to incorporate the additional fidelity, diversity, and memorization metrics. In addition, the entire subsection on **Evaluation Metrics** has been thoroughly rewritten and updated (see Page 19 in the revised manuscript). For the reviewer's convenience, we reproduce the updated subsection below:

3. Evaluation Metrics

To measure the performance of the model, we organize our evaluation framework into three categories: (i) fidelity, (ii) diversity, and (iii) memorization.

3.1 Fidelity

Fidelity reflects how closely the generated data resemble the ground truth. We first evaluate fidelity using univariate and multivariate distributional distances. The first metric is defined by ρ_1 as follows:

$$\rho_1 = \sum_{i \in \mathcal{A}} D_{js}\left(P\left(x_i\right) \| Q\left(x_i\right)\right) + \sum_{j \in \mathcal{B}} W\left(u_j, v_j\right), \tag{19}$$

where $\mathcal{A}$ is the set of discrete variables and $\mathcal{B}$ is the set of continuous variables. For discrete variables, we use Jensen-Shannon Divergence (Lin, 2002), i.e., $D_{js}(\cdot)$, to measure the distribution of samples $P(x_i)$ and the ground truth distribution $Q(x_i)$. For continuous variables, we use the Wasserstein distance (Arjovsky et al., 2017), i.e., $W(\cdot)$, to measure the distribution of variables. The distribution of the Wasserstein distance is as follows:

$$W(u, v) = \int_{-\infty}^{+\infty} |U(x) - V(x)|\, dx, \tag{20}$$

where $U(x)$ and $V(x)$ are the estimated CDFs of the respective distributions u and v. The metric ρ_1 computes the mean performance of the univariate distributions, and then contrasts this average across various models.

For the multivariate distribution, we consider the following variables in Table 4. These variables are derived from multiple factors; for instance, driving speed is calculated by dividing the travel distance by the driving duration, and charging power is determined by the difference between the end SoC and start SoC divided by the charging duration. For discrete variables such as user label, geographic index, and battery capacity, we first group the data, and then compare the continuous variables across these groups. This analysis includes, for instance, the spatiotemporal distributions of vehicle speed, charging power, and driving events.

This approach allows us to transform the assessment of multivariate distributions into meaningful univariate distributions. Accordingly, ρ_2 can be defined as follows:

$$\rho_2 = \sum_{i \in \mathcal{C}} D_{js}\left(P(x_i) \parallel Q(x_i)\right) + \sum_{j \in \mathcal{D}} W(u_j, v_j), \tag{21}$$

where $\mathcal{C}$ is the set of derived continuous variables and $\mathcal{D}$ is the set of derived discrete variables.

To further strengthen the analysis of mixed-type joint distributions, we have incorporated the k-nearest neighbors (kNN) based KL-divergence estimator, which has been widely adopted for discrete–continuous data (Ross, 2014; Gao et al., 2017; Zhao and Lai, 2020). Compared to kernel density estimation (KDE)–based approaches (Moon et al., 2017) and kernel mean embedding methods such as maximum mean discrepancy (MMD) (Gretton et al., 2012), the kNN estimator is computationally more scalable and, in our experiments, feasible for large-scale datasets.

The key idea is to decompose the joint distribution $P(X_d, X_c)$ of discrete variables X_d and continuous variables X_c into conditional components:

$$D_{KL}(P\|Q) = \sum_{x_d} P(x_d)\, D_{KL}\big(P(X_c \mid X_d = x_d) \,\|\, Q(X_c \mid X_d = x_d)\big) + D_{KL}(P(X_d)\|Q(X_d)),] \tag{22}$$

where P denotes the empirical distribution of the real dataset and Q denotes that of the synthetic dataset. The first term evaluates the conditional continuous distributions given each discrete configuration, and the second compares the discrete marginals.

For the discrete part, we apply Dirichlet-smoothed frequency estimates (Bishop and Nasrabadi, 2006; Murphy, 2013):

$$P(x_d) = \frac{n_{x_d} + \alpha}{N_{all} + \alpha K_s}, \tag{23}$$

where n_{x_d} is the count of subgroup x_d, N_{all} is the total number of samples, K_s the number of subgroups, and α the smoothing parameter.

For the conditional continuous part, we estimate $D_{KL}\big(P(X_c \mid x_d) \,\|\, Q(X_c \mid x_d)\big)$ within each subgroup x_d using the kNN estimator proposed by Wang et al. (2009), which has since been widely adopted for multivariate distributions (Singh and Poczos, 2016; Noshad et al., 2017; Ryu et al., 2022; Czy˙z et al., 2023):

$$\widehat{D}_{KL}(P\|Q) = d_c \cdot \frac{1}{N_r} \sum_{i=1}^{N_r} \log \frac{\nu_i}{\psi_i} + \log \frac{N_g}{N_r - 1}, \tag{24}$$

where d_c is the dimension of X_c, N_g is the number of samples drawn from P, N_r is the number of samples drawn from Q, ψ_i is the distance from x_i to its k-th nearest neighbor in P, and ν_i is the corresponding distance in Q.

In addition, following prior work on precision–recall based evaluation for generative models (Kynkaanniemi et al., 2019; Naeem et al., 2020; Cheema and Urner, 2023), we also incorporate the kNN based precision metric as a complementary fidelity measure. Precision quantifies the proportion of generated samples that lie within the support of the real data distribution. Formally,

$$\text{Precision} \;=\; \frac{1}{N_g} \sum_{i=1}^{N_g} \mathbf{1}(\exists\, r_j \in R : d(g_i, r_j) \le d_k(r_j))\,, \tag{25}$$

where $G = \{g_i\}_{i=1}^{N_g}$ denotes the set of generated samples, $R = \{r_j\}_{j=1}^{N_r}$ is the set of real samples, and $d_k(r_j)$ is the distance from r_j to its k-th nearest neighbor in R. The distance $d(g_i, r_j)$ between a generated sample g_i and a real sample r_j is defined for mixed continuous–discrete features as

$$d(g_i, r_j) \;=\; \sqrt{\sum_{a \in \mathcal{A}} \big(g_{i,a} - r_{j,a}\big)^2 \;+\; \sum_{b \in \mathcal{B}} \mathbf{1}\big(g_{i,b} \neq r_{j,b}\big)}\,, \tag{26}$$

where $\mathcal{A}$ denotes the set of continuous features and $\mathcal{B}$ the set of discrete features. For discrete attributes, similarity is assessed using the indicator function $\mathbf{1}(g_{i,b} \neq r_{j,b})$, which equals 1 if the two samples differ on discrete features and 0 otherwise. Taken together, this mixed-type distance enables the precision metric to complement ρ_1, ρ_2, and the kNN–KL divergence by providing a direct notion of fidelity grounded in local neighborhood support.

3.2 Diversity

Diversity captures the extent to which the generated data cover the support of the real data distribution, thereby reflecting the model's ability to reproduce the full variety of real-world patterns rather than concentrating on only the most common ones. To quantify diversity, we incorporate two complementary kNN based measures: recall and coverage.

Recall measures the proportion of real samples that lie within the support of the generated distribution, and is therefore complementary to precision. Formally, following Kynkaanniemiet al. (2019), Naeem et al. (2020), and Cheema and Urner (2023), recall is defined as

$$\text{Recall} = \frac{1}{N_r} \sum_{j=1}^{N_r} \mathbf{1}(\exists\, g_i \in G : d(r_j, g_i) \leq d_k(g_i)) \,, \tag{27}$$

where $d_k(g_i)$ is the distance from g_i to its k-th nearest neighbor in G. A high recall value indicates that the generated data cover a large portion of the real data space, including less frequent patterns.

While recall counts whether each real sample is supported by any generated sample, coverage measures how many distinct real samples are captured within the k-nearest-neighbor balls of generated samples. In other words, coverage emphasizes whether the generator is able to represent different modes of the real distribution. Formally, coverage is defined as:

$$\text{Coverage} = \frac{1}{N_r} \sum_{j=1}^{N_r} \mathbf{1}(\exists\, g_i \in G : d(g_i, r_j) \leq d_k(r_j)) \,, \tag{28}$$

where $d_k(r_j)$ is the distance from r_j to its k-th nearest neighbor in R. This metric highlights whether the generator reproduces the diversity of modes present in the real dataset.

3.3 Memorization

The third aspect evaluates whether the generative model memorizes the training data instead of learning generalizable patterns. The key idea, which has been widely adopted in the literature (van den Burg and Williams, 2021; Fang et al., 2024; Shi et al., 2025), is to use the closest distance ratio between training, validation, and generated samples to assess whether the model has leaked training data. Following (van den Burg and Williams, 2021), we partition the dataset into a training set R_t, a validation set R_v, and a generated set G.

For each training sample $r_t \in R_t$, we compute its nearest-neighbor distance to both the validation set and the generated set. The memorization score ρ_3 is defined as

$$\rho_3 = \frac{1}{|R_t|} \sum_{r_t \in R_t} \frac{\min_{r_v \in R_v} d(r_t, r_v)}{\min_{g \in G} d(r_t, g)}, \tag{29}$$

where $d(\cdot, \cdot)$ denotes the same mixed-type distance defined in Eq. 28. Intuitively, if generated samples set G are unnaturally close to the training samples R_t compared to the validation samples R_v, the ratio ρ_3 will exceed 1, indicating potential memorization. Conversely, if $\rho_3 < 1$, the generated data are farther from R_t than from R_v, suggesting underfitting. An ideal model achieves $\rho_3 \approx 1$, which indicates a balance between memorization and generalization. Together with other metrics, this measure completes our evaluation framework for generative models.

Reference list:

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Reviewer Comment 1.4

> While both charging and driving events are modeled, the paper does not address waiting times or queuing at public charging stations, which are common in real-world scenarios. How does the model account for or plan to incorporate this aspect?

Response

Thank you for raising this important point. We acknowledge that our current framework does not model explicit queuing because the on-board dataset does not record pre-plug waiting time or queue length at public chargers; hence, this quantity cannot be recovered ex post.

However, our framework can easily incorporate queuing in a plug-and-play manner by adding a *single* feature to the event schema:

$$\text{waiting_time} = \begin{cases} 0, & \text{if event_type} = \text{driving}, \\ w, & \text{if event_type} = \text{charging}, \end{cases} \qquad \text{with } w \geq 0.$$

This design mirrors our existing 'distance' feature (which is 0 for charging events and non-negative for driving events). Implementation-wise, we only add one output head and reuse the existing constraint mechanism used for waiting_time. To address this point, we have explicitly outlined the planned extension in the Conclusion (Page 38 of the revised manuscript).

"Another promising direction is to enrich the event schema with additional physically grounded features, such as battery degradation (state-of-health dynamics) and pre-charging waiting time at public stations, which would allow the model to capture long-term performance decay and congestion effects more realistically."

Reviewer Comment 1.5

> Battery capacities are treated as static inherent features. How does the model handle battery degradation and performance variations under extreme environmental conditions, such as very high or low temperatures?

Response

We sincerely thank the reviewer for raising this important point regarding battery degradation and performance variations under extreme environmental conditions. Our framework is fully data-driven, meaning that the model learns usage patterns directly from the data. As noted earlier, if ambient temperature (or related covariates) is included as an input feature, the model can capture shifts in energy consumption conditional on temperature, thereby adapting to both high and low temperature operating conditions. This highlights the flexibility of our framework to be extended to such scenarios.

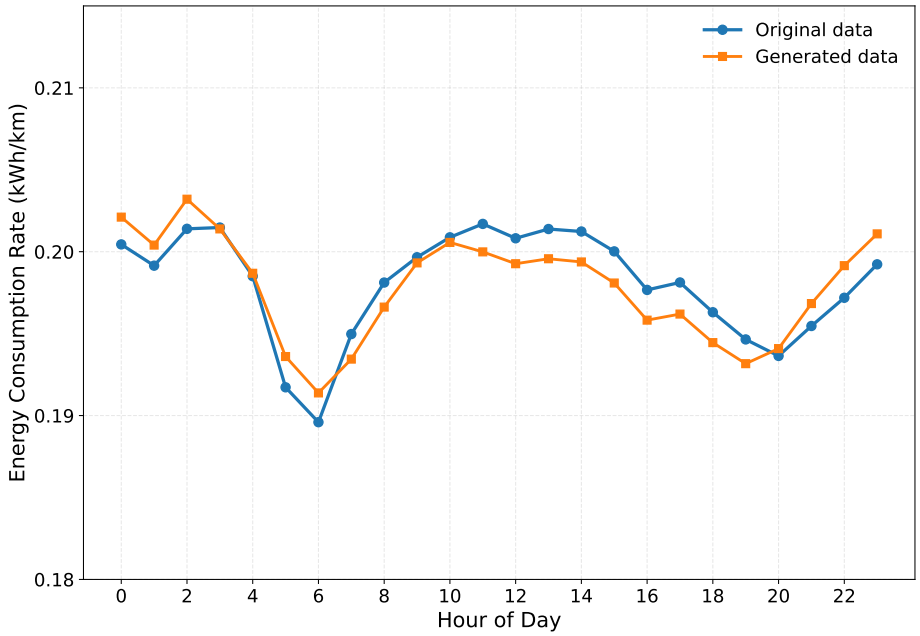

Figure A1: Comparison of average hourly energy consumption rate between original and generated data

Even without explicitly modeling degradation, we examined 24-hour (Figure A1) variations and seasonal (Figure A2) in energy consumption rate. In our previous study (Li et al., 2023), we showed that both seasonal and diurnal variations in energy consumption are closely associated

with ambient temperature. Consistent with that finding, Figure A2 compares the distributions of energy consumption rate (kWh/km) across different seasons, while Figure A1 illustrates the hourly consumption patterns within a day. From the figures, we find that the energy consumption distributions across different seasons and the consumption trends over the 24-hour cycle are largely consistent between the original and generated data. This indicates that our model is indeed able to capture and reproduce certain environmental and temporal differences.

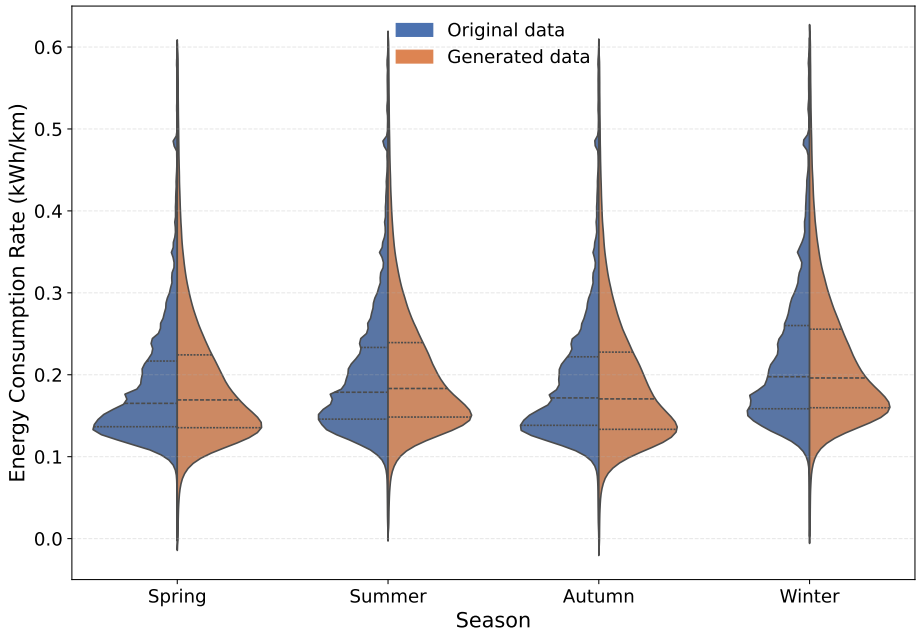

Figure A2: Comparison of seasonal energy consumption rate between original and generated data

Regarding battery degradation due to aging, since our dataset spans only about 16 months and does not include detailed state-of-health indicators, the model's ability to capture long-term degradation effects is inherently restricted. We acknowledge that battery degradation is indeed a limitation of our current work, which has been noted in the conclusion of the manuscript, and we believe that if such features were available, our framework could be naturally extended to capture degradation-related consumption patterns as well. To incorporate such limitations, similarly, we have explicitly outlined the planned extension in the Conclusion (Page 38 of the revised manuscript).

"Another promising direction is to enrich the event schema with additional physically grounded features, such as battery degradation (state-of-health dynamics) and pre-charging waiting time at public stations, which would allow the model to capture long-term performance decay and congestion effects more realistically."

Reference list:

- Li, Z., Xu, Z., Chen, Z., Xie, C., Chen, G., & Zhong, M. (2023). An empirical analysis of electric vehicles' charging patterns. Transportation Research Part D: Transport and Environment, 117, 103651.

Reviewer Comment 1.6

> Why is "end SoC" a dynamic generative feature rather than a dynamic iterative feature?

Response

Thanks for the comment. The end SoC is treated as a *generative feature* rather than an iterative one because it represents the event-level outcome to be synthesized. Given the start SoC of a driving or charging event, the model generates the corresponding end SoC, which directly reflects how much energy is consumed or replenished during that event. Once generated, this end SoC naturally becomes the start SoC of the subsequent event through state propagation. In other words, the *generation* happens within the event (start $\rightarrow$ end), while the *iteration* occurs implicitly across events when the generated end SoC is carried over to the next event. This design cleanly separates event-level generation from sequence-level state propagation.

Reviewer Comment 1.7

> On page 21, there is a typo: "Li et al. (2023),," contains an extra comma.

Response

We thank the reviewer for pointing out this typographical error. The extra comma in "Li et al. (2023)," on page 21 (original manuscript) has been corrected on Page 24 (updated manuscript), and we have carefully checked the revised version for similar issues.

Reviewer Comment 1.8

> In Figure 5, the metrics ρ_1 and ρ_2 are summed to determine the optimal transformer model layer. Since these metrics may have different units, does summing them directly provide a meaningful measure?

Response

We thank the reviewer for raising this important point. You are correct that directly summing ρ_1 and ρ_2 may not provide a meaningful measure, as the metrics differ in units and scales. Therefore, we chose not to use their summation and instead adopted a radar chart (see Figure A3) in the revised manuscript. Specifically, in line with the reviewer's earlier suggestion, we expanded our evaluation methodology to include additional fidelity, diversity, and memorization metrics, and re-ran the experiments accordingly. The updated results are presented in Table A3 below.

Table A3: Comparison of results for different transformer parameter configurations

Layer	d_{model}	Fidelity				Diversity		Memorization	Param. Size (M)
		ρ_1	ρ_2	Joint-KL	Precision	Recall	Coverage	ρ_3	
16	64	0.47	0.66	1.03	0.66	0.82	0.79	0.77	1.15
32	64	0.38	0.55	0.99	0.69	0.91	0.81	0.80	1.95
48	64	0.21	0.51	0.95	0.73	0.92	0.82	0.85	2.75
64	64	**0.17**	0.49	**0.92**	0.78	**0.97**	**0.84**	0.88	3.55
16	128	0.46	0.67	1.05	0.65	0.80	0.80	0.75	3.91
32	128	0.23	0.50	0.95	0.76	0.94	0.82	0.85	7.08
48	128	0.18	**0.47**	0.92	**0.80**	0.95	0.82	**0.89**	10.26
64	128	0.21	0.48	0.98	0.76	0.93	0.81	0.87	13.43

As the revised results indicate, the main conclusion remains unchanged. The results confirm that depth is the primary driver of performance improvements: increasing layers consistently reduces errors and boosts recall and coverage, while width contributes only moderate and sometimes inconsistent gains, mainly in precision and memorization. The 64-layer, $d_{\text{model}} = 64$ configuration stands out as particularly efficient, achieving low errors, high recall, and strong memorization with only about 3.5 million parameters. Wider and deeper variants, such as the 48-layer, $d_{\text{model}} = 128$, provide relatively balanced performance across all metrics, but at the expense of substantially larger parameter counts.

To provide a more comprehensive visualization, we additionally present radar charts (Figure A3) comparing all configurations.

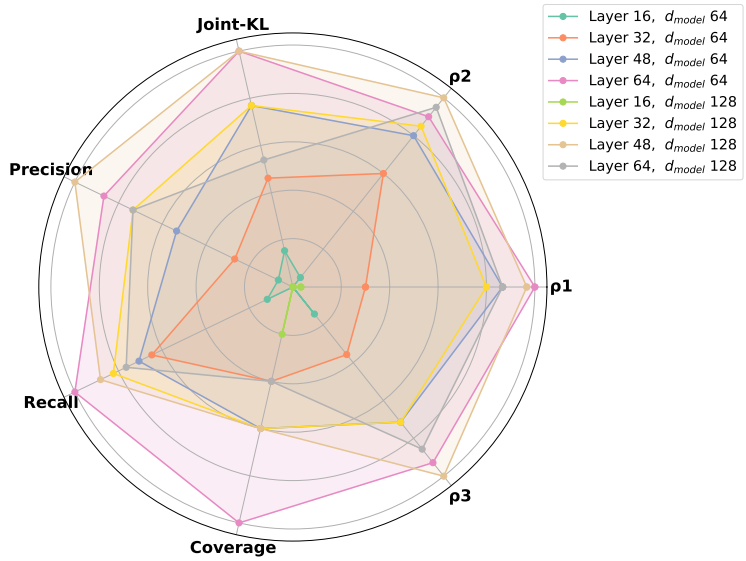

Figure A3: Comparison of Transformer depth and width configurations

Since the metrics differ in both scale and optimization direction, all values are normalized to the range $[0, 1]$ prior to plotting. For error-type (smaller-is-better) measures, we apply an inverted

min–max transformation:

$$s = 1 - \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \tag{A1}$$

whereas for larger-is-better measures (Precision, Recall, Coverage, ρ_3), we use the standard min–max scaling:

$$s = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \tag{A2}$$

Similarly, we have updated the manuscript to explicitly reflect this extension. The revised discussion has been added on Page 25 of the revised manuscript, where the updated Figure 5 corresponds to our Figure A3.

" *We evaluated transformer models by systematically varying depth (number of layers) and width (embedding size, d_{model}), using 8-head attention. As summarized in Table 6, performance improvements are primarily driven by depth: adding layers consistently reduces marginal and joint errors while enhancing recall and coverage, whereas widening the embedding dimension provides only modest and sometimes inconsistent gains, mainly in precision and memorization. To capture these trends more holistically, we visualize the results with radar charts (see Figure 5), where each axis corresponds to one of the evaluation metrics—fidelity (ρ_1, ρ_2, Joint-KL, Precision), diversity (Recall, Coverage), and memorization (ρ_3). Since the metrics differ in scale and optimization direction, they are normalized to $[0, 1]$, with error-type measures inverted so that larger values uniformly indicate better performance. The radar plots show that the 64-layer, $d_{model} = 64$ configuration already spans much of the performance space, achieving low errors, high recall, and strong memorization with only about 3.5M parameters. This demonstrates that deep but moderately wide transformers can approximate the performance of much larger models while avoiding the steep computational costs of width scaling.* "

Reviewer Comment 1.9

> Table 6 shows that all Transformer models have 64 layers, whereas all LSTM models have only 24 layers. Using fewer layers for LSTM models may negatively impact their performance, potentially biasing the comparison. Please comment.

Response

We thank the reviewer for this constructive comment. You are correct that the use of 24-layer LSTM models in our original experiments may have limited their representational capacity compared to the 64-layer Transformer models, potentially biasing the comparison. In our initial setup, we adopted 24-layer LSTMs with a hidden size of 128 to keep the overall parameter count comparable to that of the 64-layer Transformer models. However, we acknowledge that depth plays a crucial role in model performance, and a shallower LSTM configuration may not provide a fair basis for comparison.

To address this concern, we re-ran the experiments by increasing the depth of the LSTM models to 64 layers, while slightly enlarging their hidden dimension so that the overall parameter sizes remain comparable to the Transformer configurations. In addition, following the reviewer's earlier

suggestion, we also incorporated additional evaluation metrics beyond ρ_1 and ρ_2 to provide a more comprehensive assessment of model performance. The revised results, summarized in Table A4,

Table A4: Comparison of results for different generative model configurations

Model	Layer	d_{model}	Fidelity				Diversity		Memorization	Param. Size (M)
			ρ_1	ρ_2	Joint-KL	Precision	Recall	Coverage	ρ_3	
Transformer+Gibbs	64	64	0.17	**0.49**	**0.92**	**0.78**	**0.97**	0.84	**0.88**	3.55
Transformer+WGAN	64	64	**0.12**	1.04	1.18	0.67	0.83	**0.89**	0.76	3.49
Transformer+Bayes	64	64	0.92	0.97	1.21	0.71	0.65	0.78	0.67	3.37
Transformer+VAE	64	64	0.81	2.06	1.35	0.63	0.95	0.79	0.70	6.74
LSTM+Gibbs	64	80	0.20	0.57	0.97	0.73	0.92	0.82	0.80	3.67
LSTM+WGAN	64	80	0.14	1.06	1.21	0.68	0.80	0.85	0.75	3.60
LSTM+Bayes	64	80	1.09	1.02	1.25	0.69	0.59	0.75	0.62	3.47
LSTM+VAE	64	80	0.83	2.10	1.38	0.61	0.93	0.78	0.62	6.92

The revised results, summarized in Table A4, indicate that our main conclusions remain unchanged. The Transformer+Gibbs configuration continues to deliver the most balanced performance, while LSTM-based models, even at increased depth, consistently lag behind. Across architectures, Transformer-based models generally outperform their LSTM counterparts, with systematic advantages in fidelity, diversity, and memorization. The superiority of Transformers is particularly evident in Gibbs-based configurations, where they achieve lower joint errors and higher memorization, suggesting a more faithful approximation of the underlying data-generating process. Even within WGAN-, Bayes-, and VAE-based settings, Transformers tend to deliver comparable or superior recall and precision, demonstrating greater robustness across different generative paradigms.

Since Table A4 (corresponding to Table 7 in the revised manuscript) was revised to reflect the updated experiments, the corresponding discussion in the manuscript has also been updated accordingly. The modified text appears on Page 26 of the revised manuscript as follows:

" *As shown in Table 7, the four generative methods exhibit distinct behaviors across fidelity, diversity, and memorization. In terms of fidelity, Gibbs-based models emerge as the most balanced, combining relatively low joint errors with consistently higher precision, indicating that they capture both marginal characteristics and inter-variable dependencies more effectively than alternative approaches. WGAN models demonstrate strong performance on single-variable fidelity but encounter difficulties when modeling higher-order relationships, resulting in diminished overall precision and a tendency toward partial mode collapse. Bayes-based models achieve moderate levels of precision yet fall short in capturing both marginal and joint structures, reflecting their conservative generative nature. VAE-based models perform least effectively, with marked challenges in representing multivariate dependencies and consequently the lowest precision among the methods considered.*

With respect to diversity, Gibbs-based approaches again provide the most favorable balance, achieving high recall together with stable coverage, thereby reflecting both comprehensive exploration of the support of the real distribution and concentration around realistic samples. WGAN variants emphasize coverage, generating samples that remain close to the true distribution but at the expense of recall, consistent with incomplete representation of less frequent or peripheral patterns. VAE models display the inverse pattern: their samples exhibit broader dispersion, which leads to high recall but comparatively weaker coverage and fidelity, indicative of diffuse and less accurate reconstructions. Bayes-based methods remain the most restric-

tive, with consistently lower recall, underscoring their tendency to generate a narrower set of patterns. In terms of memorization, Gibbs models demonstrate the strongest alignment with training data distributions, followed by WGAN, whereas Bayes and VAE variants consistently lag behind.

Across architectures, Transformer-based models generally outperform their LSTM counterparts, with systematic advantages in fidelity, diversity, and memorization. The superiority of Transformers is particularly evident in Gibbs-based configurations, where they achieve lower joint errors and higher memorization, suggesting a more faithful approximation of the underlying data-generating process. Even within WGAN-, Bayes-, and VAE-based settings, Transformers tend to deliver comparable or superior recall and precision, demonstrating greater robustness across different generative paradigms. "

Response To Reviewer #2

> This paper proposes a novel generative framework combining a decoder-only Transformer, Gibbs sampling, and Gaussian Mixture Models to synthesize joint electric vehicle (EV) driving and charging event data. The model is trained on 15 months of real-world EV data from Shanghai and incorporates dynamic spatiotemporal features and user heterogeneity. Overall, it is a good work. I have some concerns as well as suggestions for authors consideration.

We sincerely thank the reviewer for the constructive comments and valuable suggestions. We have carefully addressed all concerns, and our detailed responses are provided below point by point.

Reviewer Comment 2.1

> The paper does not rigorously evaluate the tractability and convergence behavior of the Gibbs sampler in the high-dimensional, mixed-type feature space. As the number of features grows, sampling can become computationally expensive and prone to slow mixing, particularly in the presence of strong inter-feature dependencies or multimodal conditionals. The assumption of conditional tractability may break down, affecting sample fidelity. The authors must include convergence diagnostics and analyze the impact of feature update ordering and the number of Gibbs iterations on generation quality.

Response

We sincerely thank the reviewer for pointing out the importance of rigorously evaluating the tractability and convergence behavior of the Gibbs sampler in the high-dimensional, mixed-type feature space. We agree that Gibbs sampling may become computationally expensive and potentially suffer from slow mixing in the presence of strong inter-feature dependencies or multimodal conditionals.

In fact, we have carefully considered both the efficiency and the sampling quality of the Gibbs procedure during our model development. Our empirical findings show that a simple sequential-update Gibbs scheme is already highly effective: convergence is typically reached within about 15 iterations, and improvements from more sophisticated sampling strategies were found to be limited. To further address the reviewer's concern, we conducted a systematic convergence diagnostic experiment as follows:

- We first trained a fixed generative model and then performed Gibbs sampling with random feature update ordering.
- For each Gibbs iteration, we computed the posterior log-likelihood, i.e., the sum of log-probabilities of all features under their respective conditional models.

- We repeated the procedure over 100 runs, each with a batch size of 1024 samples and a temporal length of 60 time steps, and reported both the mean and standard deviation of the log-likelihood across runs.

The results (Figure A4) show that the posterior log-likelihood converges rapidly within 15 steps, with variance across different update orders remaining negligible. Since our generative space involves only about seven key features, two full sweeps (about 14 iterations) are sufficient for practical convergence. Even if the feature dimensionality were to increase (e.g., up to 100 features), the procedure remains scalable, as batch-based implementations can efficiently parallelize thousands of samples, provided memory or GPU capacity is sufficient.

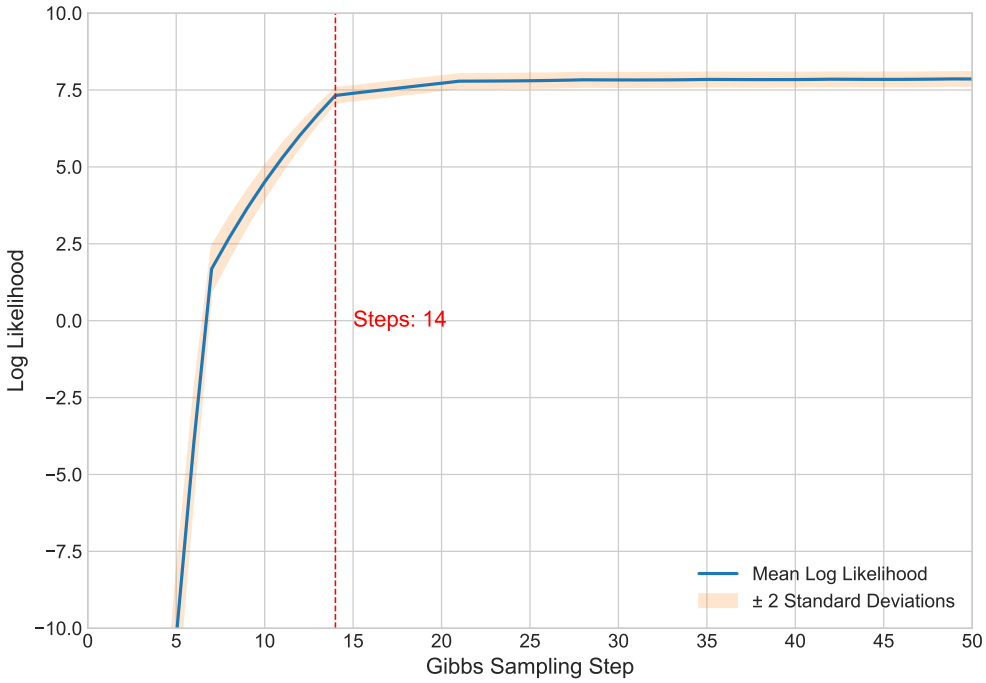

Figure A4: Convergence behavior of the Gibbs sampler

Finally, we note that in our application domain, real-time generation is not required; instead, generation fidelity and stability are prioritized over raw sampling speed. Nevertheless, computational efficiency remains an important aspect, and in future work we plan to explore accelerated sampling strategies to further reduce runtime without sacrificing sample fidelity.

To enhance the completeness of the discussion, we explicitly analyze the convergence speed of Gibbs sampling and its implications for scalability. The corresponding section appears on Page 32 of the revised manuscript, Section 4.5, and Figure 8 of the revised manuscript corresponds to Figure A4 in this letter. The revised text is reproduced below for clarity:

"*We further investigate the convergence behavior of the proposed Gibbs sampling procedure. Specifically, we trained a fixed Transformer+Gibbs model (64 layers, d_{model}=64) and generated samples over 100 batches (batch size = 1024, time steps = 60). At each Gibbs iteration, we computed the posterior log-likelihood by summing the conditional log-probabilities across all features. As shown in Figure 13, the posterior log-likelihood converges rapidly within approximately 15 iterations, with negligible variance across different*

feature update orders. These results indicate that Gibbs sampling reaches stable convergence with relatively few updates, and that update ordering is not a critical factor for our task. Since our generative space currently involves only seven key features, two complete sweeps ($\approx$ 14 iterations) are sufficient in practice. Looking ahead, when the feature dimensionality grows, more efficient variants such as block Gibbs updates or parallel tempering could be employed to further improve mixing speed and scalability."

Reviewer Comment 2.2

> The model encodes logical constraints via penalty terms, but it remains unclear how often generated samples violate domain-specific physical constraints (e.g., negative SoC, charging distance > 0). An evaluation of constraint satisfaction rates in the synthetic data is necessary to validate the reliability of the generation under real-world feasibility.

Response

We thank the reviewer for this valuable suggestion. To explicitly assess constraint satisfaction, we evaluated the generated data against domain-specific physical rules (e.g., SoC within [0,1], monotonic SoC dynamics during driving/charging, and zero distance during charging)As noted in Section 2.2 of the original manuscript (Page 16, last paragraph), we further applied a resampling procedure during generation to enforce physical consistency at the event level. With this mechanism in place, constraint satisfaction exceeds 99.95% across approximately 7 million synthetic samples. The remaining violations ($< 0.05\%$) stem mainly from minor numerical imprecision during the resampling step. Overall, the framework thus ensures strong physical fidelity, with only negligible deviations observed. To report this result clearly, we have added an explicit statement in the revised manuscript on Page 25 (Section 4.2). The added sentence reads as follows:

"Across all generative experiments, constraint satisfaction exceeds 99.95%. The rare violations ($< 0.05\%$), which stem from numerical imprecision in the resampling step."

Reviewer Comment 2.3

> The proposed framework relies on extensive high-resolution longitudinal data for effective training, raising concerns about scalability in data-scarce or cold-start environments. A sensitivity analysis on input data volume (e.g. reduced sample size) and spatial coverage would enhance understanding of the model's generalization potential.

Response

We appreciate the reviewer's suggestion and have accordingly conducted additional sensitivity analyses to evaluate the model's robustness under reduced data availability. Specifically, we reduced the input data volume in two different ways:

1. **Training set size**: we obtained proportionally reduced datasets by randomly sampling the same percentage of vehicle owners (5%, 10%, 20%, 30%, and 50%).

2. **Spatial coverage**: because random 5% subsamples already cover nearly 100% of spatial locations (as individual vehicles typically traverse multiple regions), it is infeasible to limit spatial extent simply by random sampling. Instead, we adopted a search-based procedure: starting with users whose one-year travel footprint covered the fewest spatial locations, we gradually added other users so that the marginal increase in spatial coverage was minimized. We then reported results at spatial coverage thresholds of 20%, 40%, 60%, and 80%. It should be noted, however, that although the coverage percentages appear large, the corresponding sample sizes remained below 4% of the original dataset.

Table A5: Sensitivity analysis under different training set sizes and spatial coverage

Factor	Fraction	Fidelity				Diversity		Memorization
		ρ_1	ρ_2	Joint-KL	Precision	Recall	Coverage	ρ_3
Training set size	5%	0.40	0.62	1.10	0.62	0.86	0.88	0.73
	10%	0.35	0.56	1.04	0.65	0.88	0.86	0.79
	20%	0.26	0.58	1.01	0.85	0.94	0.89	0.82
	30%	0.21	0.53	0.98	0.74	0.96	0.85	0.84
	50%	0.21	0.51	0.99	0.75	0.96	0.82	0.84
	100%	0.17	0.49	0.91	0.78	0.97	0.84	0.88
Spatial coverage	20%	0.48	0.78	1.28	0.59	0.88	0.89	0.68
	40%	0.45	0.75	1.21	0.61	0.89	0.85	0.67
	60%	0.45	0.74	1.21	0.61	0.85	0.85	0.68
	80%	0.44	0.74	1.20	0.60	0.87	0.84	0.68

Table A5 presents the sensitivity analysis under different data-scarcity conditions. For the **Training set size** dimension, the results show that even with limited data the framework can capture basic patterns, although the overall fidelity remains low. As the proportion of training users increases, performance improves steadily across all metrics (ρ_1, ρ_2, Precision, Recall, and Coverage), and the results become progressively more stable. Notably, when using 30%–50% of the dataset, the model already achieves performance close to the full-sample case, suggesting that reliable outcomes can be obtained without requiring the entire dataset.

In contrast, the **spatial coverage** experiments show less encouraging results. This is because, even though the spatial coverage ratios range from 20% to 80%, the corresponding sample sizes remain extremely small (all below 4%, with the minimum below 1%), which limits the overall learning capacity of the model. Nevertheless, since these subsets are more spatially concentrated, the generated samples can still capture a relatively high proportion of the covered regions, leading to recall and coverage values that remain at comparatively high levels despite the limited training size.

Since the experiments based on spatial coverage do not provide substantial insights due to the extremely small effective sample sizes (all below 4%), we only incorporated the **sample size**

sensitivity analysis into the revised manuscript. The updated text is included on Page 34 of the revised version, with the corresponding results shown in Table A5 in this letter, which corresponds to Table 10 of the revised manuscript. For the reviewer's convenience, the newly added text is reproduced below:

" *To further examine the robustness of our framework, we conduct a sensitivity analysis by varying the proportion of vehicle owners included in the training dataset. Specifically, we randomly sampled the same proportion of users (5%, 10%, 20%, 30%, and 50%) to construct reduced datasets and compared the results with the full-sample case. The evaluation metrics are reported in Table 10.*

The results indicate that model performance improves substantially as the sample size increases. In particular, precision, recall, and coverage show steady gains from 5% to 20%, and the outcomes with 30% and 50% of users already approximate the results obtained with the full dataset. These results indicate that even with very limited data, our framework can still capture certain usage patterns, although the precision remains relatively low. Nevertheless, coverage and recall are reasonably maintained, reflecting that the model can still reproduce the diversity of behaviors. As the training set size increases, the learned patterns become more accurate: precision improves substantially, while coverage and recall also continue to increase. At around 30%–50% of the dataset, the model achieves a favorable balance across precision, recall, and coverage, yielding performance close to the full-sample case."

Reviewer Comment 2.4

> While the paper evaluates joint distributions via derived variables (e.g., speed, charging power), the visualizations (e.g., Figure 7) show clear deviations in some subgroups. These differences need to be quantified systematically using additional multivariate similarity measures (e.g., KL-divergence over multidimensional feature space) beyond aggregated metrics, especially to evaluate conditional dependencies.

Response

We thank the reviewer for this insightful suggestion. First, let us clarify that our original ρ_2 metric aggregates multiple variables (e.g., duration, distance, SoC) into derived indicators (e.g., speed, charging power) and then evaluates the alignment of their empirical distributions across categories using Wasserstein distance. This procedure effectively transforms a multivariate joint distribution problem into comparisons over one-dimensional derived marginals, which provides interpretive value. Nevertheless, the reviewer's concern is well taken: how to directly evaluate mixed-type joint distributions (involving both discrete and continuous variables) is indeed a challenging in the literature.

To address this, we have incorporated the k-nearest neighbors (kNN) based KL-divergence estimator, which has been widely adopted for tabular data with mixed discrete–continuous features (see, e.g., Ross, 2014; Gao et al., 2017; Zhao & Lai, 2020). Compared to kernel density estimation (KDE)–based approaches (Moon et al., 2017) and kernel mean embedding methods such as maximum mean discrepancy (MMD) (Gretton et al., 2012), the kNN estimator is computationally more scalable and, in our experiments, was feasible for large-scale datasets.

The key idea is to decompose the joint distribution $P(X_d, X_c)$ of discrete variables X_d and continuous variables X_c into conditional components:

$$D_{KL}(P\|Q) = \sum_{x_d} P(x_d)\, D_{KL}\big(P(X_c \mid X_d = x_d) \,\|\, Q(X_c \mid X_d = x_d)\big) + D_{KL}(P(X_d)\|Q(X_d)), \quad \text{(A3)}$$

where P denotes the empirical distribution of the real dataset and Q denotes that of the synthetic dataset. Here, the first term evaluates the conditional continuous distributions given each discrete configuration, and the second term compares the discrete marginals.

For the discrete part, we apply Dirichlet-smoothed frequency estimates (see, e.g., Bishop, 2006; Murphy, 2012):

$$P(x_d) = \frac{n_{x_d} + \alpha}{N + \alpha K}, \quad \text{(A4)}$$

where n_{x_d} is the count of subgroup x_d, K the number of subgroups, and α the smoothing parameter.

For the conditional continuous part, we estimate $D_{KL}\big(P(X_c \mid x_d) \,\|\, Q(X_c \mid x_d)\big)$ within each subgroup x_d using the k-nearest neighbor (kNN) estimator proposed by Wang et al. (2009), which has since been widely adopted for estimating KL divergence between multivariate continuous distributions (see, e.g., Singh et al., 2016; Noshad et al., 2017; Ryu et al., 2022; Czyż et al., 2023).

$$\widehat{D}_{KL}(P\|Q) = d \cdot \frac{1}{n} \sum_{i=1}^{n} \log \frac{\nu_i}{\psi_i} + \log \frac{m}{n-1}, \quad \text{(A5)}$$

where d is the dimension of X_c, n is the number of samples drawn from P, m is the number of samples drawn from Q, ψ_i is the distance from sample x_i to its k-th nearest neighbor in P, and ν_i is the corresponding distance in Q.

In response to the reviewer's comments, we have incorporated the latest evaluation metrics into the revised manuscript. Specifically, the updates have been made in Table 6 (Page 25, updated), Table 7 (Page 26, updated), Table 9 (Page 34, updated), and Table 11 (Page 36, updated). For the reviewer's convenience, the updated Table 7 (Comparison of results for different generative model configurations) is reproduced below (Table A6).

Table A6: Comparison of results for different generative model configurations

Model	Layer	d_{model}	Fidelity				Diversity		Memorization	Param. Size (M)
			ρ_1	ρ_2	Joint-KL	Precision	Recall	Coverage	ρ_3	
Transformer+Gibbs	64	64	0.17	**0.49**	**0.92**	**0.78**	**0.97**	0.84	**0.88**	3.55
Transformer+WGAN	64	64	**0.12**	1.04	1.18	0.67	0.83	**0.89**	0.76	3.49
Transformer+Bayes	64	64	0.92	0.97	1.21	0.71	0.65	0.78	0.67	3.37
Transformer+VAE	64	64	0.81	2.06	1.35	0.63	0.95	0.79	0.70	6.74
LSTM+Gibbs	64	80	0.20	0.57	0.97	0.73	0.92	0.82	0.80	3.67
LSTM+WGAN	64	80	0.14	1.06	1.21	0.68	0.80	0.85	0.75	3.60
LSTM+Bayes	64	80	1.09	1.02	1.25	0.69	0.59	0.75	0.62	3.47
LSTM+VAE	64	80	0.83	2.10	1.38	0.61	0.93	0.78	0.62	6.92

The corresponding discussion has also been revised, as reproduced below, and can be found on Page 26 of the revised manuscript:

" *As shown in Table 7, the four generative methods exhibit distinct behaviors across fidelity, diversity, and memorization. In terms of fidelity, Gibbs-based models emerge as the most balanced, combining relatively low joint errors with consistently higher precision, indicating that they capture both marginal characteristics and inter-variable dependencies more effectively than alternative approaches. WGAN models demonstrate strong performance on single-variable fidelity but encounter difficulties when modeling higher-order relationships, resulting in diminished overall precision and a tendency toward partial mode collapse. Bayes-based models achieve moderate levels of precision yet fall short in capturing both marginal and joint structures, reflecting their conservative generative nature. VAE-based models perform least effectively, with marked challenges in representing multivariate dependencies and consequently the lowest precision among the methods considered.*

With respect to diversity, Gibbs-based approaches again provide the most favorable balance, achieving high recall together with stable coverage, thereby reflecting both comprehensive exploration of the support of the real distribution and concentration around realistic samples. WGAN variants emphasize coverage, generating samples that remain close to the true distribution but at the expense of recall, consistent with incomplete representation of less frequent or peripheral patterns. VAE models display the inverse pattern: their samples exhibit broader dispersion, which leads to high recall but comparatively weaker coverage and fidelity, indicative of diffuse and less accurate reconstructions. Bayes-based methods remain the most restrictive, with consistently lower recall, underscoring their tendency to generate a narrower set of patterns. In terms of memorization, Gibbs models demonstrate the strongest alignment with training data distributions, followed by WGAN, whereas Bayes and VAE variants consistently lag behind.

Across architectures, Transformer-based models generally outperform their LSTM counterparts, with systematic advantages in fidelity, diversity, and memorization. The superiority of Transformers is particu-

larly evident in Gibbs-based configurations, where they achieve lower joint errors and higher memorization, suggesting a more faithful approximation of the underlying data-generating process. Even within WGAN-, Bayes-, and VAE-based settings, Transformers tend to deliver comparable or superior recall and precision, demonstrating greater robustness across different generative paradigms. "

In addition, we have thoroughly revised the **Evaluation Metrics** section of the manuscript (see Page 19 in the revised manuscript) in light of the reviewer's suggestions and recent literature. The entire section has been updated to provide a more comprehensive framework encompassing fidelity, diversity, and memorization. Notably, this revision already incorporates the updates requested in the following comment, so that all related clarifications are consolidated here for consistency. For the reviewer's convenience, the updated text is reproduced below:

3. Evaluation Metrics

To measure the performance of the model, we organize our evaluation framework into three categories: (i) fidelity, (ii) diversity, and (iii) memorization.

3.1 Fidelity

Fidelity reflects how closely the generated data resemble the ground truth. We first evaluate fidelity using univariate and multivariate distributional distances. The first metric is defined by ρ_1 as follows:

$$\rho_1 = \sum_{i \in \mathcal{A}} D_{js}\left(P\left(x_i\right) \| Q\left(x_i\right)\right) + \sum_{j \in \mathcal{B}} W\left(u_j, v_j\right), \tag{19}$$

where $\mathcal{A}$ is the set of discrete variables and $\mathcal{B}$ is the set of continuous variables. For discrete variables, we use Jensen-Shannon Divergence (Lin, 2002), i.e., $D_{js}(\cdot)$, to measure the distribution of samples $P(x_i)$ and the ground truth distribution $Q(x_i)$. For continuous variables, we use the Wasserstein distance (Arjovsky et al., 2017), i.e., $W(\cdot)$, to measure the distribution of variables. The distribution of the Wasserstein distance is as follows:

$$W(u, v) = \int_{-\infty}^{+\infty} |U(x) - V(x)| \, dx, \tag{20}$$

where $U(x)$ and $V(x)$ are the estimated CDFs of the respective distributions u and v. The metric ρ_1 computes the mean performance of the univariate distributions, and then contrasts this average across various models.

For the multivariate distribution, we consider the following variables in Table 4. These variables are derived from multiple factors; for instance, driving speed is calculated by dividing the travel distance by the driving duration, and charging power is determined by the difference between the end SoC and start SoC divided by the charging duration. For discrete variables such as user label, geographic index, and battery capacity, we first group the data, and then compare the continuous variables across these groups. This analysis includes, for instance, the spatiotemporal distributions of vehicle speed, charging power, and driving events.

This approach allows us to transform the assessment of multivariate distributions into meaningful univariate distributions. Accordingly, ρ_2 can be defined as follows:

$$\rho_2 = \sum_{i \in \mathcal{C}} D_{js}\left(P(x_i) \parallel Q(x_i)\right) + \sum_{j \in \mathcal{D}} W(u_j, v_j), \tag{21}$$

where $\mathcal{C}$ is the set of derived continuous variables and $\mathcal{D}$ is the set of derived discrete variables.

To further strengthen the analysis of mixed-type joint distributions, we have incorporated the k-nearest neighbors (kNN) based KL-divergence estimator, which has been widely adopted for discrete–continuous data (Ross, 2014; Gao et al., 2017; Zhao and Lai, 2020). Compared to kernel density estimation (KDE)–based approaches (Moon et al., 2017) and kernel mean embedding methods such as maximum mean discrepancy (MMD) (Gretton et al., 2012), the kNN estimator is computationally more scalable and, in our experiments, feasible for large-scale datasets.

The key idea is to decompose the joint distribution $P(X_d, X_c)$ of discrete variables X_d and continuous variables X_c into conditional components:

$$D_{KL}(P\|Q) = \sum_{x_d} P(x_d)\, D_{KL}\big(P(X_c \mid X_d = x_d) \,\|\, Q(X_c \mid X_d = x_d)\big) + D_{KL}(P(X_d)\|Q(X_d)),] \quad (22)$$

where P denotes the empirical distribution of the real dataset and Q denotes that of the synthetic dataset. The first term evaluates the conditional continuous distributions given each discrete configuration, and the second compares the discrete marginals.

For the discrete part, we apply Dirichlet-smoothed frequency estimates (Bishop and Nasrabadi, 2006; Murphy, 2013):

$$P(x_d) = \frac{n_{x_d} + \alpha}{N_{all} + \alpha K_s}, \quad (23)$$

where n_{x_d} is the count of subgroup x_d, N_{all} is the total number of samples, K_s the number of subgroups, and α the smoothing parameter.

For the conditional continuous part, we estimate $D_{KL}\big(P(X_c \mid x_d) \,\|\, Q(X_c \mid x_d)\big)$ within each subgroup x_d using the kNN estimator proposed by Wang et al. (2009), which has since been widely adopted for multivariate distributions (Singh and Poczos, 2016; Noshad et al., 2017; Ryu et al., 2022; Czy˙z et al., 2023):

$$\widehat{D}_{KL}(P\|Q) = d_c \cdot \frac{1}{N_r} \sum_{i=1}^{N_r} \log \frac{\nu_i}{\psi_i} + \log \frac{N_g}{N_r - 1}, \quad (24)$$

where d_c is the dimension of X_c, N_g is the number of samples drawn from P, N_r is the number of samples drawn from Q, ψ_i is the distance from x_i to its k-th nearest neighbor in P, and ν_i is the corresponding distance in Q.

In addition, following prior work on precision–recall based evaluation for generative models (Kynkaanniemi et al., 2019; Naeem et al., 2020; Cheema and Urner, 2023), we also incorporate the kNN based precision metric as a complementary fidelity measure. Precision quantifies the proportion of generated samples that lie within the support of the real data distribution. Formally,

$$\text{Precision} \;=\; \frac{1}{N_g} \sum_{i=1}^{N_g} \mathbf{1}(\exists\, r_j \in R : d(g_i, r_j) \le d_k(r_j))\,, \quad (25)$$

where $G = \{g_i\}_{i=1}^{N_g}$ denotes the set of generated samples, $R = \{r_j\}_{j=1}^{N_r}$ is the set of real samples, and $d_k(r_j)$ is the distance from r_j to its k-th nearest neighbor in R. The distance $d(g_i, r_j)$ between a generated sample g_i and a real sample r_j is defined for mixed continuous–discrete features as

$$d(g_i, r_j) \;=\; \sqrt{\sum_{a \in \mathcal{A}} \big(g_{i,a} - r_{j,a}\big)^2 \;+\; \sum_{b \in \mathcal{B}} \mathbf{1}\big(g_{i,b} \neq r_{j,b}\big)}\,, \quad (26)$$

where $\mathcal{A}$ denotes the set of continuous features and $\mathcal{B}$ the set of discrete features. For discrete attributes, similarity is assessed using the indicator function $\mathbf{1}(g_{i,b} \neq r_{j,b})$, which equals 1 if the two samples differ on discrete features and 0 otherwise. Taken together, this mixed-type distance enables the precision metric to complement ρ_1, ρ_2, and the kNN–KL divergence by providing a direct notion of fidelity grounded in local neighborhood support.

3.2 Diversity

Diversity captures the extent to which the generated data cover the support of the real data distribution, thereby reflecting the model's ability to reproduce the full variety of real-world patterns rather than concentrating on only the most common ones. To quantify diversity, we incorporate two complementary kNN based measures: recall and coverage.

Recall measures the proportion of real samples that lie within the support of the generated distribution, and is therefore complementary to precision. Formally, following Kynkaanniemiet al. (2019), Naeem et al. (2020), and Cheema and Urner (2023), recall is defined as

$$\text{Recall} = \frac{1}{N_r} \sum_{j=1}^{N_r} \mathbf{1}(\exists\, g_i \in G : d(r_j, g_i) \leq d_k(g_i)), \tag{27}$$

where $d_k(g_i)$ is the distance from g_i to its k-th nearest neighbor in G. A high recall value indicates that the generated data cover a large portion of the real data space, including less frequent patterns.

While recall counts whether each real sample is supported by any generated sample, coverage measures how many distinct real samples are captured within the k-nearest-neighbor balls of generated samples. In other words, coverage emphasizes whether the generator is able to represent different modes of the real distribution. Formally, coverage is defined as:

$$\text{Coverage} = \frac{1}{N_r} \sum_{j=1}^{N_r} \mathbf{1}(\exists\, g_i \in G : d(g_i, r_j) \leq d_k(r_j)), \tag{28}$$

where $d_k(r_j)$ is the distance from r_j to its k-th nearest neighbor in R. This metric highlights whether the generator reproduces the diversity of modes present in the real dataset.

3.3 Memorization

The third aspect evaluates whether the generative model memorizes the training data instead of learning generalizable patterns. The key idea, which has been widely adopted in the literature (van den Burg and Williams, 2021; Fang et al., 2024; Shi et al., 2025), is to use the closest distance ratio between training, validation, and generated samples to assess whether the model has leaked training data. Following (van den Burg and Williams, 2021), we partition the dataset into a training set R_t, a validation set R_v, and a generated set G.

For each training sample $r_t \in R_t$, we compute its nearest-neighbor distance to both the validation set and the generated set. The memorization score ρ_3 is defined as

$$\rho_3 = \frac{1}{|R_t|} \sum_{r_t \in R_t} \frac{\min_{r_v \in R_v} d(r_t, r_v)}{\min_{g \in G} d(r_t, g)}, \tag{29}$$

where $d(\cdot,\cdot)$ denotes the same mixed-type distance defined in Eq. 28. Intuitively, if generated samples set G are unnaturally close to the training samples R_t compared to the validation samples R_v, the ratio ρ_3 will exceed 1, indicating potential memorization. Conversely, if $\rho_3 < 1$, the generated data are farther from R_t than from R_v, suggesting underfitting. An ideal model achieves $\rho_3 \approx 1$, which indicates a balance between memorization and generalization. Together with other metrics, this measure completes our evaluation framework for generative models.

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Reviewer Comment 2.5

> Although the ρ_3 metric aims to measure memorization, its reliance on nearest-neighbor distances can be sensitive to scaling and data sparsity. Authors should additionally report precision-recall or coverage metrics (e.g., train-sample overlap rate, novelty rate) to support claims of generalization.

Response

We sincerely thank the reviewer for raising this important point. Our original evaluation metric, ρ_3, follows prior literature in using the closest distance ratio to measure the risk of data leakage (van den Burg & Williams, 2021; Fang et al., 2024). As the reviewer correctly points out, however, additional metrics such as precision, recall, and coverage provide complementary perspectives. While ρ_3 focuses on memorization, these measures better reflect the balance between sample fidelity and diversity. We fully agree that they are reasonable indicators of model performance, and therefore we have incorporated them into our evaluation framework to strengthen the reliability of our conclusions.

We carefully reviewed prior work on precision–recall based metrics. Early approaches defined the neighborhood radius using a fixed threshold ϵ (Sajjadi et al., 2018), which can be sensitive to scaling and density variations. More recent work has demonstrated that k-nearest-neighbor (kNN) based definitions are more robust and adaptive to local data density (Cheema & Urner, 2023; Kynkäänniemi et al., 2019; Naeem et al., 2020). In particular, Naeem et al. (2020) proposed reliable fidelity and diversity metrics for generative models, which we adopt in this study. Specifically, we define the following kNN-based metrics:

1. **Precision (fidelity):** measures the proportion of generated samples that lie within the support of the real data distribution.

$$\text{Precision} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}(\exists\, r_j \in R : d(g_i, r_j) \le d_k(r_j))\,, \tag{A6}$$

   where $d_k(r_j)$ denotes the distance from r_j to its k-th nearest neighbor in the real dataset R.

2. **Recall (diversity):** quantifies the proportion of real samples that are covered by the generated distribution.

$$\text{Recall} = \frac{1}{m}\sum_{j=1}^{m} \mathbf{1}(\exists\, g_i \in G : d(r_j, g_i) \le d_k(g_i))\,, \tag{A7}$$

   where $d_k(g_i)$ is the distance from g_i to its k-th nearest neighbor in the generated dataset G.

3. **Coverage (diversity):** complements recall by evaluating how well the generated samples avoid mode dropping and capture the spread of real data.

$$\text{Coverage} = \frac{1}{m}\sum_{j=1}^{m} \mathbf{1}(\exists\, g_i \in G : d(g_i, r_j) \le d_k(r_j))\,, \tag{A8}$$

Together, these metrics complement ρ_3 by providing a more balanced assessment of fidelity, diversity, and generalization in generative models.

In the same manner as our response to the previous comment, we have updated all experimental tables and the entire **Evaluation Metrics** section accordingly (see Page 19 in the revised manuscript). To avoid redundancy, we kindly refer the reviewer to our detailed explanation and the revised text already provided in the response to Comment 2.4. These updates apply consistently across the manuscript.

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